

Empirical Bayes Double Shrinkage for Combining Biased and Unbiased Causal Estimates

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Outline

- 1 Motivating Problem
- 2 Estimator Construction
- 3 Application to the WHI

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Problem in *integrating heterogeneous data*, e.g. combining RCTs with observational studies to estimate causal effects (Cheng and Cai, 2021; Shi et al., 2023; Colnet et al., 2024; Yang et al., 2024; Chen et al., 2025; Dang et al., 2025; Lin et al., 2025; van der Laan et al., 2026).

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Causal Setting: Transportability Assumptions

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- Define subgroup-specific CATEs for each population:

$$\tau_{uk} = \mathbb{E}_U[Y_i(1) - Y_i(0) \mid S_i = k]$$

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- **Transportability assumption:**

$$\tau_k \equiv \tau_{uk} = \tau_{bk}, \quad k = 1, \dots, K$$

i.e. treatment effects equal within each stratum

- Approximate validity may still yield substantial risk reduction

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Hierarchical Model

Simple model generalizing one introduced in [Green and Strawderman \(1991\)](#):

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$$\boldsymbol{\xi}_b \sim \mathcal{N}(0, \gamma^2 \mathbf{I}_K),$$

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$$\begin{aligned}\tau &\sim \mathcal{N}(0, \eta^2 \mathbf{I}_K), \\ \xi_b &\sim \mathcal{N}(0, \gamma^2 \mathbf{I}_K), \\ \hat{\tau}_u \mid \tau &\sim \mathcal{N}(\tau, \Sigma_u), \text{ and} \\ \hat{\tau}_b \mid \tau, \xi_b &\sim \mathcal{N}(\tau + \xi_b, \Sigma_b).\end{aligned}$$

for **unknown** hyperparameters η^2 and γ^2

Estimator Form

Estimator can be constructed as the **posterior mean** of τ under this model, which evaluates to

$$\psi_k(\eta^2, \gamma^2) = \underbrace{\left(\frac{\eta^2 (\gamma^2 + \sigma_{bk}^2 + \sigma_{uk}^2)}{\sigma_{uk}^2 (\gamma^2 + \sigma_{bk}^2) + \eta^2 (\gamma^2 + \sigma_{bk}^2 + \sigma_{uk}^2)} \right)}_{a_k(\eta^2, \gamma^2): \text{aggregate shrinkage toward zero}} \times \left(\underbrace{\frac{(\gamma^2 + \sigma_{bk}^2)}{\gamma^2 + \sigma_{bk}^2 + \sigma_{uk}^2}}_{\lambda_k(\eta^2, \gamma^2): \text{data-driven weight}} \hat{\tau}_{uk} + \underbrace{\frac{\sigma_{uk}^2}{\gamma^2 + \sigma_{bk}^2 + \sigma_{uk}^2}}_{1 - \lambda_k(\eta^2, \gamma^2)} \hat{\tau}_{bk} \right).$$

This is the **double-shrinkage** property: take a data-driven convex combo of $\hat{\tau}_u$ and $\hat{\tau}_b$ and then a Stein-like shrinkage toward zero.

Versions of the Estimator (I)

To construct a usable estimator, need estimates of η^2, γ^2 .

Three approaches from [Xie et al. \(2012\)](#)

Moment-Matching: Observing that

$$\begin{aligned}\mathbb{E}(\|\hat{\boldsymbol{\tau}}_{\mathbf{u}}\|_2^2) &= \text{Tr}(\boldsymbol{\Sigma}_{\mathbf{u}}) + K\eta^2, \quad \text{and} \\ \mathbb{E}(\|\hat{\boldsymbol{\tau}}_{\mathbf{b}} - \hat{\boldsymbol{\tau}}_{\mathbf{u}}\|_2^2) &= \text{Tr}(\boldsymbol{\Sigma}_{\mathbf{u}}) + \text{Tr}(\boldsymbol{\Sigma}_{\mathbf{b}}) + K\gamma^2,\end{aligned}$$

use the estimates:

$$\begin{aligned}\hat{\eta}_{\text{mm}}^2 &= \frac{1}{K} (\|\hat{\boldsymbol{\tau}}_{\mathbf{u}}\|_2^2 - \text{Tr}(\boldsymbol{\Sigma}_{\mathbf{u}}))_+ \\ \hat{\gamma}_{\text{mm}}^2 &= \frac{1}{K} (\|\hat{\boldsymbol{\tau}}_{\mathbf{u}} - \hat{\boldsymbol{\tau}}_{\mathbf{b}}\|_2^2 - \text{Tr}(\boldsymbol{\Sigma}_{\mathbf{u}}) - \text{Tr}(\boldsymbol{\Sigma}_{\mathbf{b}}))_+.\end{aligned}$$

Versions of the Estimator (II)

Maximum Likelihood: Observing that

$$\mathcal{L}(\eta^2, \gamma^2) \propto \prod_k (\eta^2 + \sigma_{uk}^2)^{-1/2} e^{-\frac{\hat{\tau}_{uk}^2}{2(\eta^2 + \sigma_{uk}^2)}} \times \\ \prod_k (\eta^2 + \gamma^2 + \sigma_{bk}^2)^{-1/2} e^{-\frac{\hat{\tau}_{bk}^2}{2(\eta^2 + \gamma^2 + \sigma_{bk}^2)}}.$$

We can numerically optimize to obtain the estimates

$$(\hat{\eta}_{\text{mle}}^2, \hat{\gamma}_{\text{mle}}^2) = \operatorname{argmax}_{\eta^2, \gamma^2 \geq 0} \log \left(\mathcal{L}(\eta^2, \gamma^2) \right).$$

Versions of the Estimator (III)

URE Minimization: We can also minimize an unbiased risk estimate (URE). Here,

$$\begin{aligned} \text{URE}(\eta^2, \gamma^2) = & \text{Tr}(\mathbf{\Sigma}_u) + \sum_k (\psi_k(\eta^2, \gamma^2) - \hat{\tau}_{uk})^2 - \\ & 2 \sum_k \sigma_{uk}^2 \cdot (1 - a_k(\eta^2, \gamma^2) \cdot \lambda_k(\eta^2, \gamma^2)). \end{aligned}$$

We can numerically optimize to obtain the estimates

$$(\hat{\eta}_{\text{ure}}^2, \hat{\gamma}_{\text{ure}}^2) = \underset{\eta^2, \gamma^2 \geq 0}{\text{argmin}} \text{URE}(\eta^2, \gamma^2).$$

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- **EB coverage** is a frequently-used weaker condition
 - Implies **average coverage**: under fixed τ , a $1 - \alpha$ fraction of effects are covered with high probability in large samples
 - However, some outlying effects may not be covered with $1 - \alpha$ probability across repeated samples of the data

Inference

- Straightforward to extend the results of [Armstrong et al. \(2020\)](#) to construct confidence intervals
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Definition (Robust EB Confidence Intervals (EBCIs))

The robust EBCI for ψ_k , the causal effect estimate obtained from any version of double-shrinkage estimators, is

$$\psi_k \pm cva(c_k) \hat{a}_k \sqrt{\left(\hat{\lambda}_k^2 \sigma_{uk}^2 + (1 - \hat{\lambda}_k)^2 \sigma_{bk}^2 \right)},$$

where $\hat{a}_k$ and $\hat{\lambda}_k$ are the shrinkage factors, and $cva(c_k)$ is an inflation factor whose form is given in [Armstrong et al. \(2020\)](#).

Outline

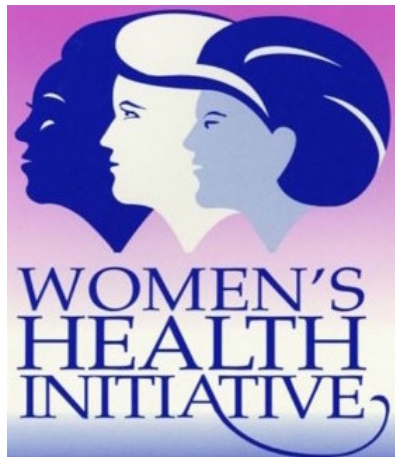
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WHI Overview

Dataset Overview

- Study of postmenopausal women initiated in 1991
- RCT of hormone therapy (HT) w/ 16k enrollees
- ODB w/ 50k comparable enrollees

Consider the effect of HT on coronary heart disease (CHD)



Results: Point Estimation

Subgroup Variable(s)	# of Strata	MSE as a % of $\hat{\tau}_u$ MSE				
		δ_1 (G&S)	δ_2 (G&S)	$\hat{\psi}_{mm}$	$\hat{\psi}_{mle}$	$\hat{\psi}_{ure}$
CVD	2	100%	100%	21%	<u>16%</u>	32%
Age	3	62%	73%	21%	<u>16%</u>	34%
Sun	5	39%	52%	11%	<u>9%</u>	15%
CVD, Age	6	40%	83%	21%	<u>21%</u>	27%
CVD, Sun	10	33%	87%	17%	<u>17%</u>	19%
Age, Sun	15	21%	44%	8%	<u>8%</u>	10%
CVD, Age, Sun	30	51%	80%	20%	<u>20%</u>	20%

Table 1: MSE reduction for different stratifications, with RCT sample size of 1,000. Samples obtained by bootstrapping from the WHI. Best estimator is underlined. ↗ ↘ ↙ ↚

Results: Average Coverage

Subgroup Variable(s)	# of Strata	$\hat{\psi}_{\text{mm}}$		$\hat{\psi}_{\text{mle}}$		$\hat{\psi}_{\text{ure}}$	
		Δ Len	Avg Covg.	Δ Len	Avg Covg.	Δ Len	Avg Covg.
CVD	2	17%	98%	17%	98%	16%	98%
Age	3	22%	100%	23%	100%	21%	99%
Sun	5	28%	100%	29%	100%	28%	100%
CVD, Age	6	29%	99%	30%	99%	29%	99%
CVD, Sun	10	34%	99%	34%	99%	33%	99%
Age, Sun	15	40%	100%	41%	100%	40%	100%
CVD, Age, Sun	30	50%	100%	50%	100%	50%	100%

Table 2: Robust CI length reductions (Δ Len) relative to Wald Intervals, and *average* coverage across strata, for different stratifications. RCT sample size of 1,000. Samples obtained by bootstrapping from the WHI.

Results: Minimum Coverage

Subgroup Variable(s)	# of Strata	$\hat{\psi}_{\text{mm}}$		$\hat{\psi}_{\text{mle}}$		$\hat{\psi}_{\text{ure}}$	
		Δ Len	Min Covg.	Δ Len	Min Covg.	Δ Len	Min Covg.
CVD	2	17%	96%	17%	97%	16%	94%
Age	3	22%	97%	23%	98%	21%	96%
Sun	5	28%	97%	29%	98%	28%	97%
CVD, Age	6	29%	96%	30%	96%	29%	95%
CVD, Sun	10	34%	97%	34%	97%	33%	96%
Age, Sun	15	40%	96%	41%	96%	40%	96%
CVD, Age, Sun	30	50%	94%	50%	92%	50%	92%

Table 3: Robust CI length reductions (Δ Len) relative to Wald Intervals, and *minimum* coverage across strata, for different stratifications. RCT sample size of 1,000. Samples obtained by bootstrapping from the WHI.

Future Work

Current paper available at [arXiv:2309.06727](https://arxiv.org/abs/2309.06727).

Future work

- Current approach underperforms when true effects are far from normality
- Various extensions with fewer parametric assumptions, making use of Tweedie's formula, NPMLE ([Efron, 2016](#); [Luo et al., 2023](#); [Kiefer and Wolfowitz, 1956](#))

References (I)

- Armstrong, T. B., Kolesár, M., and Plagborg-Møller, M. (2020). Robust empirical bayes confidence intervals. *arXiv preprint arXiv:2004.03448*.
- Chen, S., Li, S., Zhang, B., and Ye, T. (2025). Minimax rates and adaptivity in combining experimental and observational data. *Journal of Causal Inference*, 13(1):20240024.
- Cheng, D. and Cai, T. (2021). Adaptive combination of randomized and observational data. *arXiv preprint arXiv:2111.15012*.
- Colnet, B., Mayer, I., Chen, G., Dieng, A., Li, R., Varoquaux, G., Vert, J.-P., Josse, J., and Yang, S. (2024). Causal inference methods for combining randomized trials and observational studies: a review. *Statistical Science*, 39(1):165–191.
- Dang, L. E., Tarp, J. M., Abrahamsen, T. J., Kvist, K., Buse, J. B., Petersen, M., and van der Laan, M. (2025). Experiment-selector cross-validated targeted maximum likelihood estimator for hybrid rct-external data studies. *Journal of Causal Inference*, 13(1):20240041.
- Efron, B. (2016). Empirical bayes deconvolution estimates. *Biometrika*, 103(1):1–20.
- Green, E. J. and Strawderman, W. E. (1991). A James-Stein type estimator for combining unbiased and possibly biased estimators. *Journal of the American Statistical Association*, 86(416):1001–1006.
- Green, E. J., Strawderman, W. E., Amateis, R. L., and Reams, G. A. (2005). Improved estimation for multiple means with heterogeneous variances. *Forest Science*, 51(1):1–6.
- Hoff, P. and Yu, C. (2019). Exact adaptive confidence intervals for linear regression coefficients. *Electronic Journal of Statistics*, 13:94–119.
- Kiefer, J. and Wolfowitz, J. (1956). Consistency of the maximum likelihood estimator in the presence of infinitely many incidental parameters. *The Annals of Mathematical Statistics*, pages 887–906.
- Li, K.-C. et al. (1985). From Stein's unbiased risk estimates to the method of generalized cross validation. *The Annals of Statistics*, 13(4):1352–1377.
- Lin, X., Tarp, J. M., and Evans, R. J. (2025). Combining experimental and observational data through a power likelihood. *Biometrics*, 81(1):ujaf008.

References (II)

- Luo, J., Banerjee, T., Mukherjee, G., and Sun, W. (2023). Empirical bayes estimation with side information: A nonparametric integrative tweedie approach. *arXiv preprint arXiv:2308.05883*.
- Rosenman, E., Basse, G., Owen, A., and Baiocchi, M. (2020). Combining observational and experimental datasets using shrinkage estimators. *arXiv preprint arXiv:2002.06708*.
- Shi, X., Pan, Z., and Miao, W. (2023). Data integration in causal inference. *Wiley Interdisciplinary Reviews: Computational Statistics*, 15(1):e1581.
- Stein, C. (1956). Inadmissibility of the usual estimator for the mean of a multivariate normal distribution. Technical report, Stanford University Stanford United States.
- van der Laan, M., Qiu, S., Tarp, J. M., and van der Laan, L. (2026). Adaptive-tmle for the average treatment effect based on randomized controlled trial augmented with real-world data. *Journal of Causal Inference*, 14(1):20240025.
- Xie, X., Kou, S., and Brown, L. D. (2012). Sure estimates for a heteroscedastic hierarchical model. *Journal of the American Statistical Association*, 107(500):1465–1479.
- Yang, S., Liu, S., Zeng, D., and Wang, X. (2024). Data fusion methods for the heterogeneity of treatment effect and confounding function. *Bernoulli*.

Thanks!